

Unitree Week 17+18

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Updates

- Still working on Teleoperation and Imitation Learning on IsaacLab and IsaacSim
- Due to various issues with software management, I had to reinstall IsaacLab and IsaacSim.
- Tested various IsaacLab programs to verify software functionality.
- Tested pre-defined teleoperation programs on IsaacLab

Log

- Download IsaacSim 5.1.0 online
- Downloaded IsaacLab v2.3.2 Repository using command: “git clone <https://github.com/isaac-sim/IsaacLab.git> C:\Users\Mario\IsaacLab”
- Bridged IsaacSim to Isaac Lab using command: cmd /c mklink /D _isaac_sim "C:\Users\Mario\IsaacSim"
- Installed IsaacLab into the IsaacLab repository: .\isaacclab.bat --install

Log

- Errors needed to be fixed with h5py:
.\isaac_sim\python.bat -m pip install h5py==3.15.1
- Ran [demo.py](#) file:
.\isaaclab.bat -p scripts/demos/bipeds.py
- Obtained file location for SE3 teleoperation code:
“scripts/environments/teleoperation/teleop_se3_agent.py”
- Command for teleop_se3_agent with Cube-Franka:
.\isaaclab.bat -p
scripts/environments/teleoperation/teleop_se3_agent.py --task
Isaac-Lift-Cube-Franka-v0 --teleop_device keyboard

Log

- Note for removing Cache (removes previous IsaacSim setup), delete file “extscache” (gets made each time IsaacSim is used).
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Cube-Franka

The `teleop_se3_agent.py` code allows for moving the Franka Robot using keyboard buttons W,A,S,D

- This code places the robot back to its initial state after any movement.



Explaining teleop_se3_agent.py

- Code used to move the Franka Robot.
- Important Dependencies: “Pinocchio” → handles Rigid Body Dynamics.
- Teleop Interface: translates keypresses (such as “W”) into a 6 degree of freedom command (3D Position and 3D rotation).
 - Obtains latest keyboard input
 - Combines movement command with the “gripper” command
 - Sends this to the physics engine, moving the robot.

Using this code for the Unitree G1

- The primary step for using this code for the G1/R1 robots is implementing more d.o.f's onto the program.
- Enabling an Inverse Kinematics Controller allows to control the higher quantity of d.o.f's.
- Enabling pinocchio ensures the G1 knows how the joints are connected by importing locomanipulation tasks.
- Changing the code from assuming a single-arm setup to dual-arm setup.